

Let $A \in \mathbb{R}^{3 \times 3}$ such that the following holds true

$$A^5 + 2A^3 + A = 0_{3 \times 3}$$

Prove that A is diagonalisable in $\mathbb{C}$

Let C be a basis of $\mathbb{C}^3$ and G be a 3×3 matrix whose columns are the vectors of C ; such that the matrix $J = G^{-1}AG$ is in the Jordan normal form. Such basis C must exist because A necessarily has 3 eigenvalues (including algebraic multiplicity). Let J have eigenvalues of $\lambda_1, \lambda_2, \lambda_3$.

$$J = \begin{pmatrix} \lambda_1 & 1 \vee 0 & 0 \\ 0 & \lambda_2 & 1 \vee 0 \\ 0 & 0 & \lambda_3 \end{pmatrix}$$

The polynomial $A^5 + 2A^3 + A = 0$ can be therefore substituted into as

$$GJ^5G^{-1} + 2GJ^3G^{-1} + GJG^{-1} = 0_{3 \times 3}$$

Multiplying both sides by G from the right and G^{-1} from the left

$$J^5 + J^3 + J = 0_{3 \times 3}$$

Or factorised

$$J(J + iI)^2(J - iI)^2 = 0_{3 \times 3}$$

If there were a λ^* eigenvalue of A such that λ^* isn't one of the solutions (from now x_i) of the polynomial given above, the diagonal of $J - x_i I$ in the Jordan blocks belonging to λ^* would not be 0.

$$J = \begin{pmatrix} \ddots & & 0 & & 0 \\ & J_{1,\lambda^*} & \cdot & \cdot & \cdot \\ & \cdot & J_{2,\lambda^*} & \cdot & \cdot \\ 0 & \cdot & \cdot & \ddots & \cdot \\ & \cdot & \cdot & \cdot & J_{k,\lambda^*} \\ 0 & & 0 & & \ddots \end{pmatrix}$$

For J_{i,λ^*} ; $i \in 1, \dots, k$

$$J_{i,\lambda^*} - x_i I = \begin{pmatrix} \lambda^* - x_i & 1 & \cdots & 0 \\ 0 & \lambda^* - x_i & \cdots & 0 \\ \vdots & \vdots & \ddots & 1 \\ 0 & 0 & 0 & \lambda^* - x_i \end{pmatrix} \quad \lambda^* \neq x_i \implies \text{non 0 diagonal}$$

Therefore any higher power of that Jordan block wouldn't be 0 and since this would be the case for all $J - x_i I$ matrices; the product of their higher powers would also not be 0. This means that all of the eigenvalues of A must necessarily be the solutions of $x^5 + 2x^3 + x = 0$. Here are all the possible combinations

1. $x(x - i)(x + i)$
2. $x(x - i)^2$
3. $x(x + i)^2$
4. $(x - i)(x + i)^2$
5. $(x - i)^2(x + i)$

Let $P_A(x)$ be the characteristic polynomial of A . Then $\deg(P_A(x)) = 3$ and for all $P(x)$ polynomials such that $P(A) = 0_{3 \times 3}$; the roots of that polynomial contain all the eigenvalues of A . Since $P_A(x) = \det(A - xI)$, for A real; P_A has only real number coefficients $\implies$ if P_A has any imaginary solution z , then $\bar{z}$ is also a solution of P_A . From that we get the only valid characteristic polynomial

$$x(x - i)(x + i)$$

Which implies A has three different eigenvalues and so A has three different eigenvectors in $\mathbb{C}$. Therefore A is diagonalisable.
Q. E. D.